Model card: distilled, pruned DeBERTa-v3-xsmall for RTE

[Student name] · [Roll number] · CS6315E RTE Challenge

code: `https://github.com/ihemanthm/rte-challenge`

Architecture. `microsoft/deberta-v3-xsmall` encoder (12 layers, hidden 384, 6 heads) + pooler + linear 2-way head. Full fine-tuning with knowledge distillation from a DeBERTa-v3-base teacher (masked KD, below); **no** LoRA/adapters. After training, the input word-embedding is *vocabulary-pruned*: only the rows for the 62,245 kept tokens (special tokens ∪ every training-set token ∪ the 60,000 most frequent tokens of the tokenizer's unigram model) are kept; any other token id maps to `[UNK]` through a fixed lookup table.

Parameter counts.

Total parameters	45,542,402
Trainable parameters (all; full fine-tuning)	45,542,402
word embedding ($62{,}245 \times 384$)	23,902,080
transformer layers + position/LayerNorm + head	21,640,322
Token-id remap buffer (int32, 128,100 entries; not a parameter)	0.5 MB
Before pruning (same checkpoint)	70,830,722

Training data. RTE train (2,490 pairs; identical to GLUE RTE train), minus a 10% validation split grouped by premise. Teacher soft labels are out-of-fold logits of five DeBERTa-v3-base models trained on the same train set. No external NLI/RTE data, no pre-fine-tuned RTE checkpoints. Dev (GLUE RTE validation, 277) is used only for reporting.

Training hyperparameters.

Loss	$0.5\,\mathrm{CE} + 0.5\,T^2\,\mathrm{KL}$, T=2; pure CE where the teacher is wrong
Optimiser	AdamW, weight decay 0.01 (none on bias/LayerNorm), grad-clip 1.0
Learning rate	5×10^{-5} top layer, layer-wise decay 0.9 per layer down; head 1×10^{-4}
Schedule	linear warm-up 10%, then linear decay
Batch size / max length	16 / 256 tokens (premise truncated first)
Epochs	up to 15, early stopping on validation accuracy, patience 4
Seeds	13, 21, 42, 87, 100; submitted = best val. of 20 runs (seed 21, epoch 7)
Precision	fp32 weights; bf16 autocast on CUDA, fp32 elsewhere
Pruning	post-hoc; top-N chosen on train only (98% coverage)
Hardware / software	Colab T4 GPU (reruns bit-identical); PyTorch 2.14, Transformers 5.22.2

Results (GLUE RTE dev, 277).

	Params	Dev acc.
Submitted model (masked KD, seed 21, pruned)	45.5M	78.3
Same checkpoint before pruning	70.8M	78.0
Method estimate: masked KD, mean ± sd over 5 seeds	70.8M	78.3 ± 1.4
DeBERTa-v3-base baseline / teacher (5 seeds)	184.4M	84.8 ± 1.1
Instructor reference baseline	[TBD]	[TBD]

Intended use and limits. Binary entailment on short English news-style pairs. Text with tokens outside the kept vocabulary loses those tokens to `[UNK]` (98.5% of dev tokens are kept; on the 60% of dev pairs that are fully covered, outputs equal the unpruned model exactly). Weak on high-overlap non-entailments (36.8% error), numbers (31.2%), modality.

Reproduce: see README (`make train-submission submission`).